

Two players take turns taking counters from a pile. On each turn, a player takes 1 counter or 2 counters. Whoever takes the last counter wins.

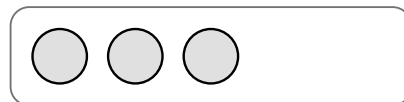
Problem 1: It is your turn. Circle the counters you take to be sure to win, or cross out the pile if you cannot be sure to win.



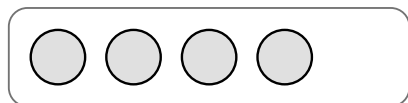
1



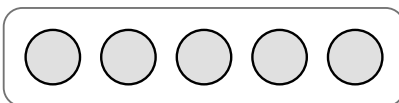
2



3

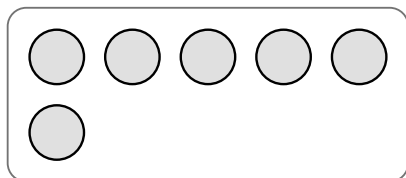


4



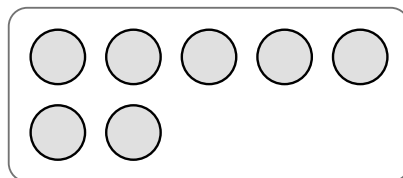
5

Problem 2: Circle 1st or 2nd to show whether you would rather go first or second with each pile.



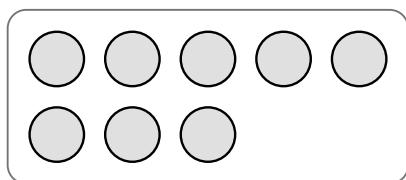
6

1st 2nd



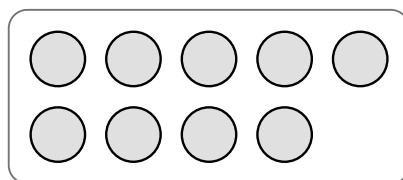
7

1st 2nd



8

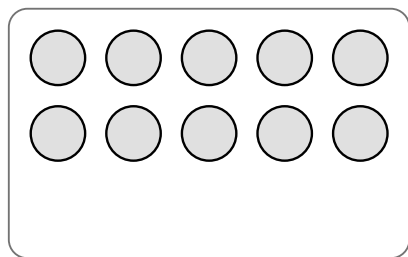
1st 2nd



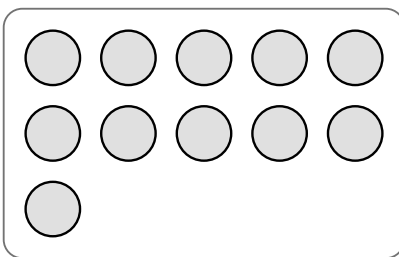
9

1st 2nd

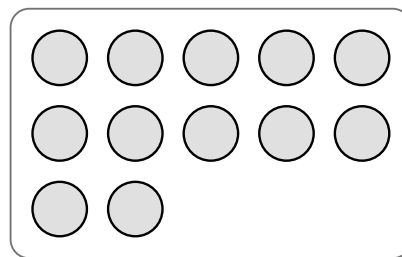
Problem 3: It is your turn with a bigger pile. Circle the counters you take to be sure to win, or cross out the pile if you cannot be sure to win.



10



11



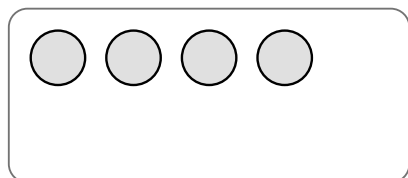
12

Problem 4: A pile is a trap if the player whose turn it is cannot be sure to win. Put an X on every trap from 1 to 20.

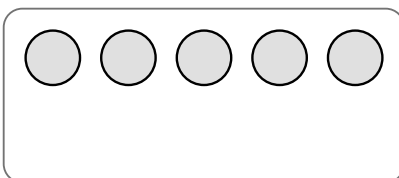
1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

11	12	13	14	15	16	17	18	19	20
----	----	----	----	----	----	----	----	----	----

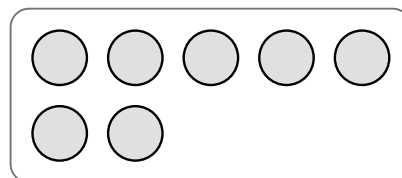
Problem 5: Ben always starts by taking 2 counters. Circle each pile where Ben can still be sure to win after that.



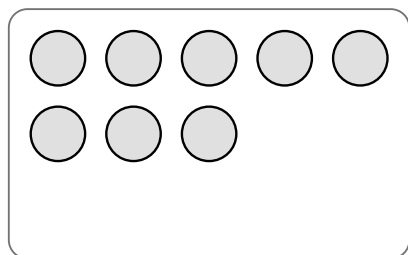
4



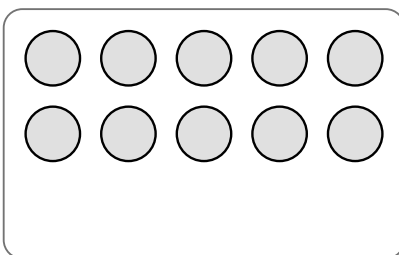
5



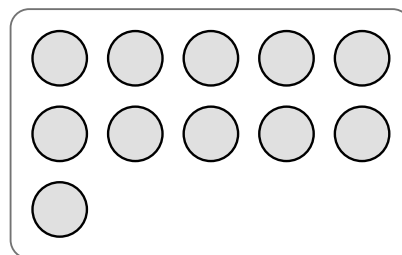
7



8

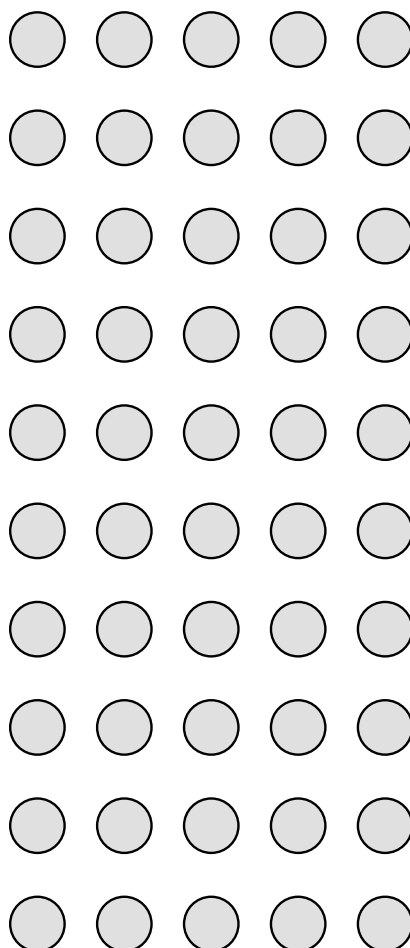
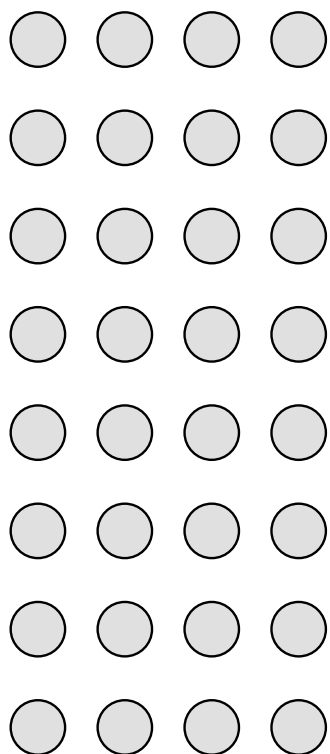


10



11

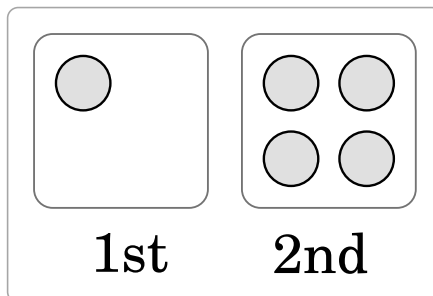
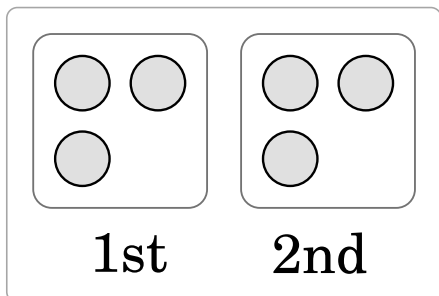
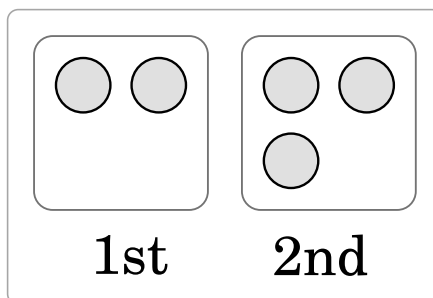
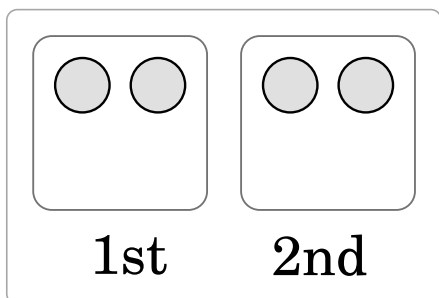
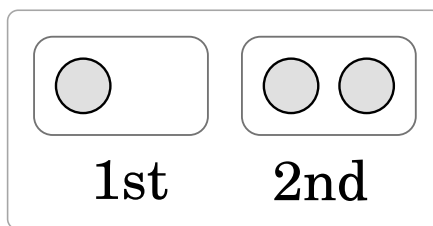
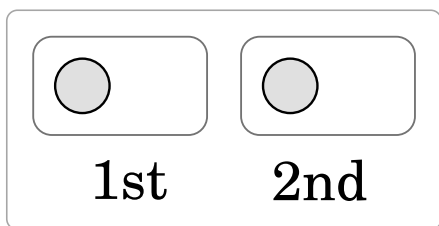
Problem 6: Show every different way a game with 4 counters can go, and then every way a game with 5 counters can go. Circle the counters taken on each turn.



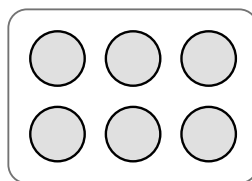
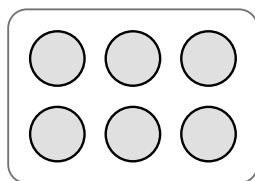
Problem 7: Now each player may take 1, 2, or 3 counters on a turn. Put an X on every trap from 1 to 20.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

Problem 8: Now there are two piles, and on your turn you take 1 or 2 counters from one pile. Circle 1st or 2nd to show whether you would rather go first or second.



Problem 9: Play the game from Problem 8 with two piles of 6 counters. If you go second, how can you win every time, whatever the other player does?



Problem 10: Now there is one pile again, and whoever takes the last counter loses. Put an X on every trap from 1 to 20.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20